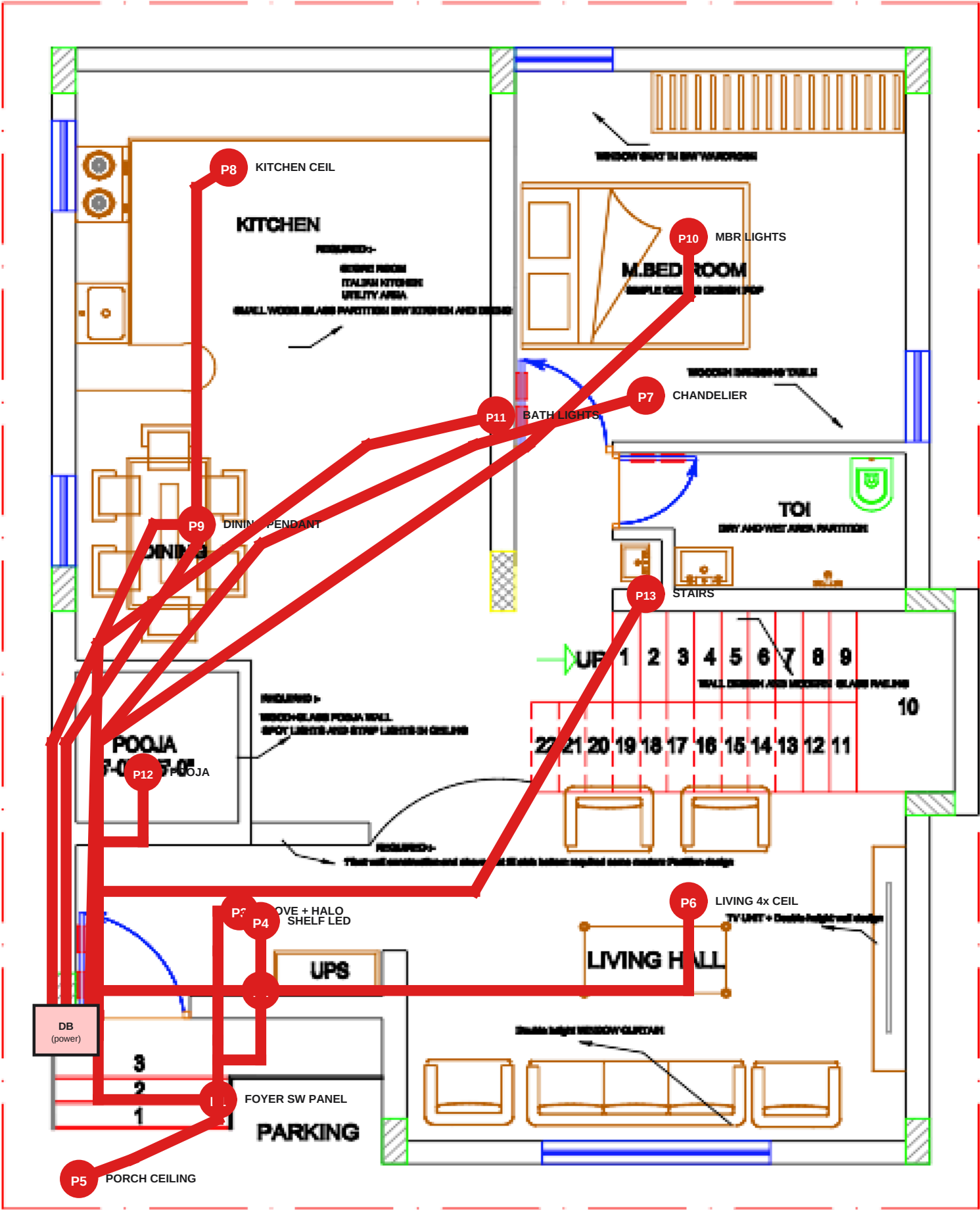




Conduit colour code

- RED - 25mm light power (1.5mm² L/N/E)
- BLUE - 25mm heavy power (2.5 / 4mm²)
- ORANGE - 25mm Cat6 / network (LV-25 grey PVC)
- PURPLE - 16mm low-voltage (LV-16)
- GREEN - Water automation (20/16mm)

Line style:

- Solid = pull cable NOW
- Dashed = pull STRING only (future)

Ref scheme:

- P = light power H = heavy power
- D = data Cat6 L = LV-16 W = water
- F = first-floor combined

Conduits on this sheet (13 runs)

Ref	Size	From -> To / What's inside / Note
P1	25mm	DB -> foyer switch panel (N wall, 18M Schneider Unica, 1200mm FFL) Inside: 1.5sqmm L + N + E (feeds 6-gang foyer panel; powers all foyer + porch lighting) 65mm-deep GI MS box. NEUTRAL must reach the panel - no exceptions.
P2	25mm	Switch panel -> 2x foyer ceiling GU10 spotlights Inside: Switched live from Gang 1 + N + E GU10 7W 2700K, adjustable gimbals to angle 30deg toward stone wall.
P3	25mm	Switch panel -> false ceiling cove driver + halo driver above cavity Inside: Switched live Gang 2 + N + E (powers cove driver + halo driver) Both drivers above false ceiling. Halo wraps cavity top, cove wraps foyer perimeter.
P4	16mm	Switch panel -> walnut shelf underneath, S wall of foyer Inside: Switched live Gang 3 + N + E (powers walnut shelf 24V driver) Floor route preferred for the wire (avoids extra ceiling penetration).
P5	16mm	Switch panel -> outside porch ceiling light Inside: Switched live Gang 5 + N + E Through N wall to porch. Outdoor IP44 fitting required.
P6	25mm	DB -> Living false ceiling, 4x recessed COB ceiling boxes (12W 2700K) Inside: 1.5sqmm L + N + E (4-way splits at false ceiling JB) Avoid the FF cut-out void area. Stay in solid-slab zone (left/west).
P7	25mm	DB -> vertical up W wall -> FF beam -> ceiling rose box at void centre Inside: 1.5sqmm L + N + E (single drop for chandelier) Terminate at ceiling rose box on FF beam level. Chandelier hangs into double-height void.
P8	25mm	DB -> kitchen ceiling box + utility light Inside: 1.5sqmm L + N + E (kitchen ceiling + utility light branch) Cool/neutral 4000K panel for task area. Utility light branched from this run.
P9	25mm	DB -> dining pendant centre + 2x supplementary downlights Inside: 1.5sqmm L + N + E (pendant + 2x downlights via JB) Pendant drop to 2100mm FFL above dining table. Confirm table position first.
P10	25mm	DB -> MBR ceiling + 2x bedside (E wall) + S/W wardrobe LED driver Inside: 1.5sqmm L + N + E (ceiling + bedside L + bedside R + wardrobe driver) Centre on bed once headboard line is marked. Bedside sconces flank headboard.
P11	25mm	DB -> bathroom ceiling + mirror light + exhaust + PIR outside door Inside: 1.5sqmm L + N + E (ceiling + mirror + exhaust + PIR switch) PIR switch outside door, W side, 1200mm FFL. Geyser switch BELOW PIR at 1050mm.
P12	25mm	DB -> Pooja ceiling (spots + decorative strip) Inside: 1.5sqmm L + N + E (Pooja spotlights + strip) Spot lights + strip lights as per the markup on AutoCAD. Confirm fixture spec on-site.
P13	25mm	DB -> step lights + mid-flight wall light + niche light + landing Inside: 1.5sqmm L + N + E (step lights + mid-flight wall light + landing) Step lights (3W each) on each riser left side. 2-way switch at top + bottom.

SCHEMATIC - verify dimensions against handwritten plan

TOP=S BOTTOM=N (main door) LEFT=E RIGHT=W



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Ref scheme:

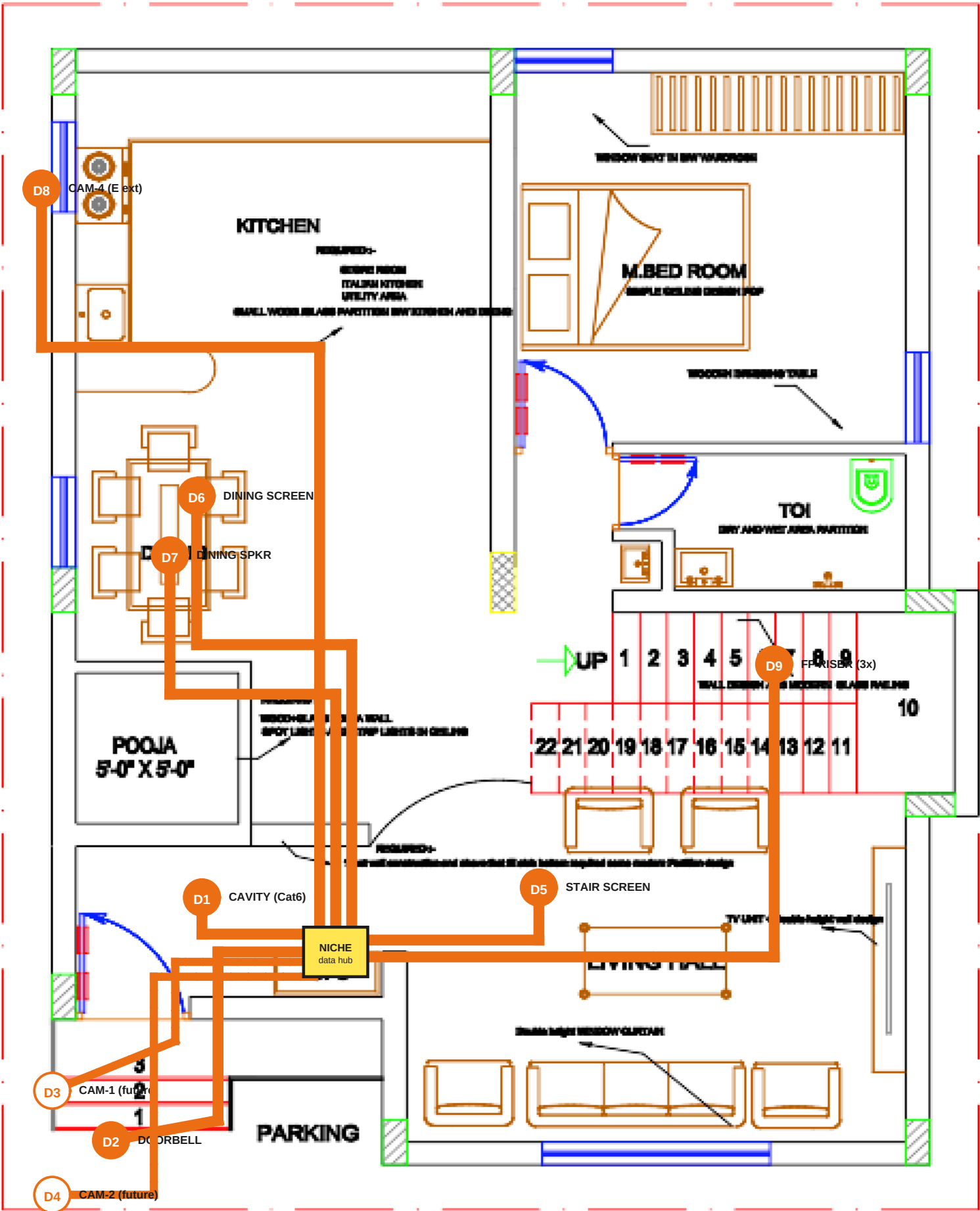
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F = first-floor combined

Conduits on this sheet (10 runs)

Ref	Size	From -> To / What's inside / Note
H1	25mm	DB -> foyer screen cavity (back wall, bottom-LEFT) Inside: 2.5sqmm L + N + E (230V to cavity 2x 5A sockets; Sonoff Mini R2 switches Monitor) <i>Cavity socket panel = 2-module on cavity BACK wall, centre 53in / 1346mm FFL.</i>
H2	25mm	DB -> TV wall (W wall of Living) - 3 boxes at 300mm FFL Inside: 2.5sqmm L + N + E - 3 boxes: 16A socket + double 5A + Cat6 keystone <i>All 3 boxes at same height. Keep Cat6 keystone box 150mm from power boxes (interference).</i>
H3	25mm	DB -> Living perimeter sockets (multiple boxes around the room) Inside: 2.5sqmm L + N + E - skirting-level run to perimeter sockets at 300mm FFL <i>Min 150mm from corners. Skirting-level chase.</i>
H4	25mm	DB -> kitchen counter wall - 4x boxes at 1100mm FFL Inside: 2.5sqmm L + N + E - 4x counter sockets (5A + 16A combo) <i>Above counter level, behind backsplash position. Confirm with kitchen designer for tile cutout.</i>
H5	25mm	DB -> chimney socket above hob at 1900mm FFL Inside: 2.5sqmm L + N + E (chimney 16A socket) <i>Centre over hob position. 16A socket.</i>
H6	25mm	DB -> hob position at 150mm FFL (floor screed route) Inside: 4sqmm L + N + E (25A hob direct hardwire) <i>DIRECT hardwire (no socket). 25A MCB. Conduit emerges from floor at hob centreline.</i>
H7	25mm	DB -> fridge niche socket at 300mm FFL Inside: 2.5sqmm L + N + E (16A fridge socket) <i>Inside fridge niche - confirm height after fridge model is picked.</i>
H8	25mm	DB -> MBR bedside boxes (E wall) at 600mm FFL + utility Inside: 2.5sqmm L + N + E (2x bedside double 5A + utility sockets) <i>Bedside sockets at 600mm FFL on E wall, flanking the headboard.</i>
H9	25mm	DB -> MBR AC socket at 1850mm FFL on N wall above door Inside: 4sqmm L + N + E (20A AC socket) <i>RCBO protection. Confirm AC manual for socket position above outdoor unit side.</i>
H10	25mm	DB -> geyser outlet at 1850mm FFL inside bathroom Inside: 2.5sqmm L + N + E (20A geyser circuit) <i>20A RCBO. Geyser SWITCH (DP) outside door at 1050mm FFL (below PIR).</i>



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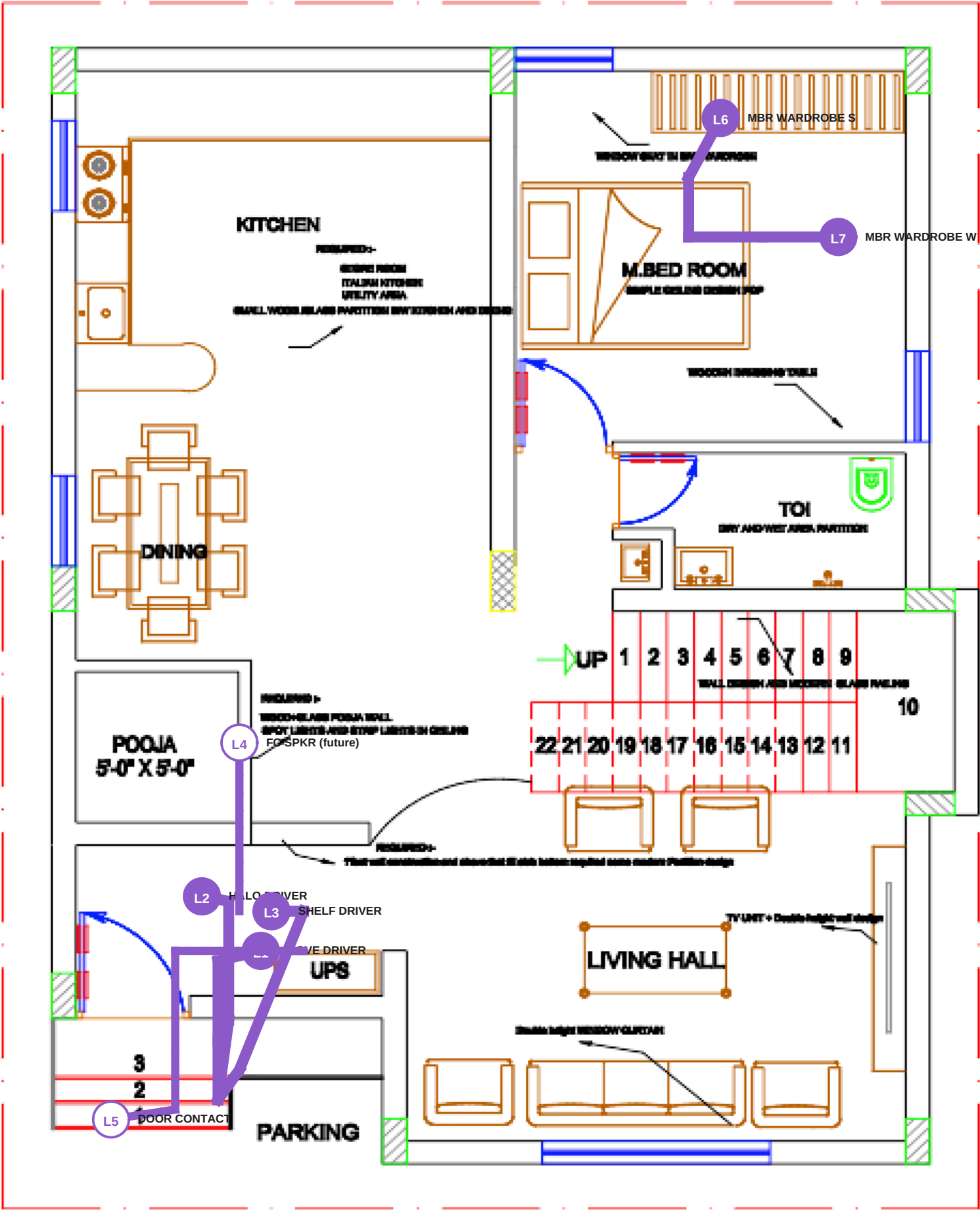
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Conduits on this sheet (9 runs)

Ref	Size	From -> To / What's inside / Note
D1	25mm	Niche -> foyer screen cavity (back wall, bottom-RIGHT) Inside: 1x indoor Cat6 (UTP) FLOOR route under screed (NOT ceiling). ~12m. Drops Cat6 to RPI behind the foyer monitor.
D2	25mm	Niche -> main-door outside face (Reolink Video Doorbell PoE) Inside: 1x outdoor LSZH UV-rated Cat6 (PoE) Floor route preferred. Doorbell at 1450mm FFL, latch side.
D3	25mm (provision)	Niche -> porch W wall (future face-detection camera) Inside: PULL STRING ONLY - capped IP67 back-box at porch W wall, 1650mm FFL DROPPED from base install (2026-05-20) - Reolink doorbell handles face capture.
D4	25mm (provision)	Niche -> porch ceiling (future overview camera, 2700mm FFL) Inside: PULL STRING ONLY - capped IP67 back-box at porch ceiling NE corner False-ceiling route only - soffit not reachable via floor.
D5	25mm	Niche -> staircase S wall, 1500mm FFL (Waveshare 10.1in panel) Inside: 1x HDMI 2.0 (3m) + 1x USB-A->micro-USB (3m) + existing Cat6 spare CUSTOM 280x195x80mm masonry cavity (not modular box). ~200mm above existing 2M switch.
D6	25mm	Niche -> dining E wall (Waveshare dining panel) Inside: 1x indoor Cat6 + 1x pull string ~12m / 35-40ft. Pi-at-screen pattern (foyer-style). Position TBD by interior designer.
D7	25mm	Niche -> ceiling JB above future dining table (mono speaker) Inside: 1x indoor Cat6 (PoE-capable) + 1x pull string Ceiling JB centred over future dining table. Foyer-pattern Pi + PAM8403 + 3in ceiling speaker.
D8	25mm	Niche -> E-wall exterior at kitchen-utility junction, 2400-2600mm FFL Inside: 1x outdoor LSZH UV Cat6 (PoE) + draw wire Horizontal run 150mm below GF slab soffit. Confirm whether utility has external door.
D9	25mm x 3	Niche -> vertical rise in staircase W wall -> FF (see FF-01 sheet) Inside: 3x LV-25 conduits up staircase W wall: (a) R-FF-1 2x Cat6, (b) R-FF-2 1x Cat6, (c) R-FF-3 1x Cat6. CAM-3+CAM-5 continue further up. Dedicated 150mm-wide chase channel. See FF-01 sheet for terminations above.



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Conduits on this sheet (7 runs)

Ref	Size	From -> To / What's inside / Note
L1	16mm	Switch panel -> cove LED driver in false ceiling Inside: 24V DC 2-core to cove driver (driver inside false ceiling perimeter) Driver sits in false ceiling perimeter. Accessible via inspection hatch.
L2	16mm	Switch panel -> halo driver above false ceiling, near cavity top Inside: 24V DC 2-core to halo driver above cavity Halo strip wraps cavity reveal. Amber 2200K. Driver in false ceiling above cavity.
L3	16mm	Switch panel -> shelf LED driver under walnut shelf Inside: 24V DC 2-core to walnut shelf 24V driver Shelf strip 6W/m, warm white. Driver under shelf carcass.
L4	16mm (provision)	Cavity TOP wall -> false ceiling above (future ceiling speaker) Inside: PULL STRING ONLY - future ceiling speaker in foyer false ceiling Conduit exits cavity TOP inner wall going up into false ceiling. Pull string only.
L5	16mm (provision)	Niche -> main door frame TOP (door contact sensor) Inside: PULL STRING ONLY - future concealed door contact sensor at top of frame Concealed sensor at top of main door frame. Future provision, no cable now.
L6	16mm	MBR ceiling JB -> S-wall wardrobe top rail LED driver Inside: 24V DC 2-core to wardrobe S-wall LED strip driver Door-activated sensor on each wardrobe leaf.
L7	16mm	MBR ceiling JB -> W-wall wardrobe tail LED driver Inside: 24V DC 2-core to wardrobe W-wall tail LED driver Tail of L-shaped wardrobe; door-activated sensor.

SCHEMATIC - verify dimensions against handwritten plan

TOP=S BOTTOM=N (main door) LEFT=E RIGHT=W



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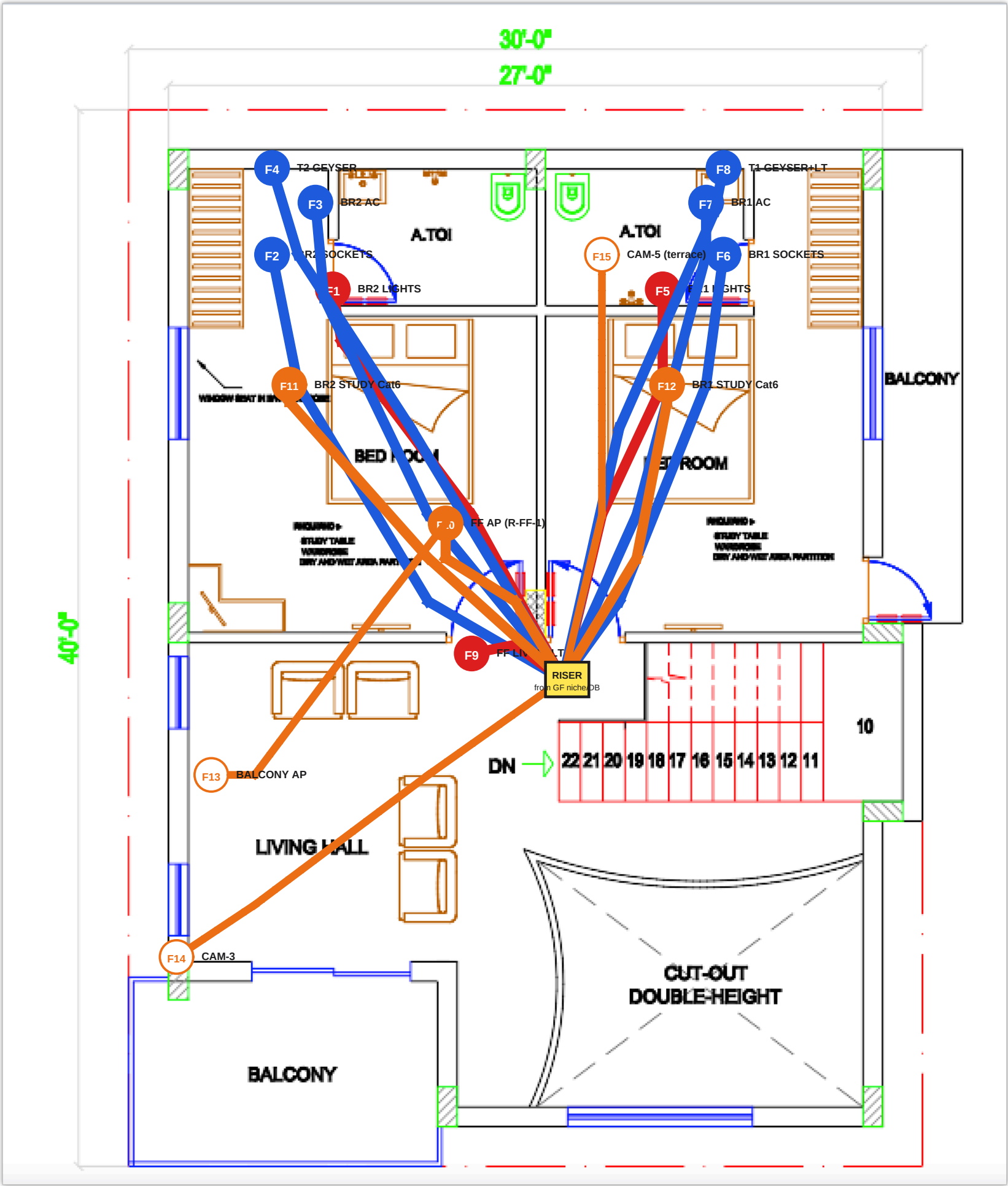
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Ref	Size	From -> To / What's inside / Note
W1	20mm	Niche -> staircase shaft -> terrace SW parapet Sintex JB (~42ft) Inside: 1x outdoor LSZH Cat6 (PoE for Sintex water-level sensor) <i>Embedded under terrace screed BEFORE tiling. IP65 JB 200x200x100mm at SW parapet.</i>
W2	16mm	Niche -> terrace Sintex float -> DB cupboard (~45ft) Inside: 2-core 1.5mm2 double-insulated (Sintex float failsafe to DB cupboard) <i>Parallel to W1, MIN 50mm separation. Terminates at float terminal block in DB cupboard.</i>
W3	20mm	Niche -> E external wall above manhole, sump JB (~25ft) Inside: 1x outdoor LSZH Cat6 (PoE for sump water-level sensor) <i>Sleeve through E wall, slope outward to prevent water ingress.</i>
W4	16mm	Niche -> sump float -> DB cupboard (~25ft) Inside: 2-core 1.5mm2 (sump float failsafe to DB cupboard) <i>Parallel to W3, MIN 50mm separation.</i>
W5 (provision)	20mm	Niche -> DB cupboard (~12ft) Inside: PULL STRING ONLY - future Cat6 backup link (water automation) <i>Both ends capped + labelled WATER AUTOMATION - FUTURE Cat6.</i>
W6	25mm	DB cupboard -> borewell head outside (confirm existing run can be repurposed) Inside: 4sqmm armoured (P1 borewell motor power, 16A Type C) <i>P1 starter sits in DB cupboard. CONFIRM existing conduit can carry new 4sqmm.</i>
W7	25mm	DB cupboard -> P2 cage east outside wall (~5ft) Inside: 2.5sqmm PVC (P2 booster pump power, 16A Type C) <i>P2 starter inside DB cupboard. Cage = lockable outdoor enclosure on east wall.</i>



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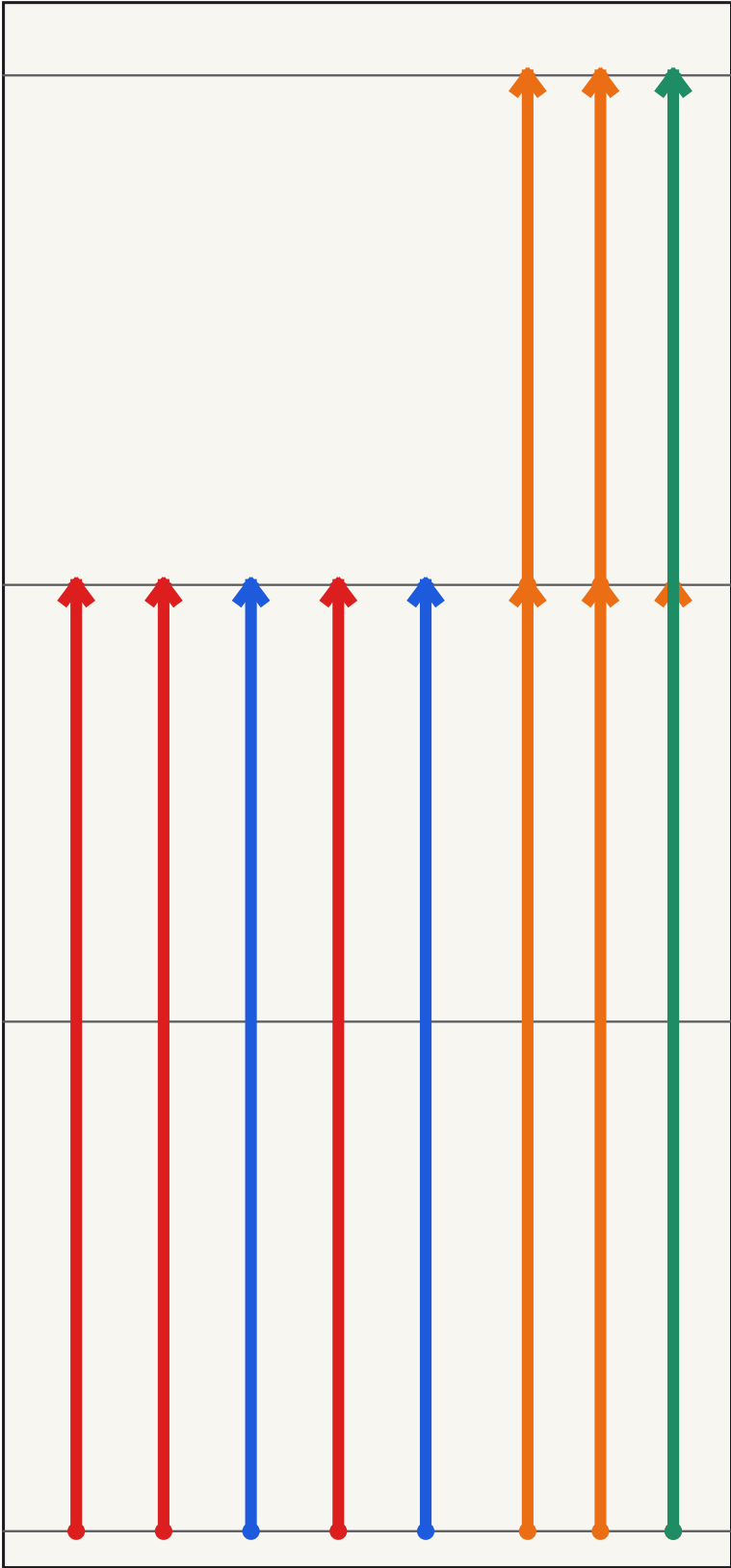
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FF conduits (15 runs)

Ref	Size	From -> To / What's inside / Note
F1	25mm	Riser -> BR2 ceiling (centred) + cove + bedsides + wardrobe driver Inside: 1.5sqmm L + N + E (BR2 ceiling + cove + bedside spots + wardrobe driver) Centre on bed once headboard line is marked.
F2	25mm	Riser -> BR2 bedside + study sockets at 600mm FFL Inside: 2.5sqmm L + N + E (bedside L + bedside R + study sockets)
F3	25mm	Riser -> BR2 AC socket at 1850mm FFL Inside: 4sqmm L + N + E (20A AC socket)
F4	25mm	Riser -> T2 geyser outlet 1850mm FFL + DP switch outside door Inside: 2.5sqmm L + N + E (20A geyser; switch outside door 1200mm FFL)
F5	25mm	Riser -> BR1 ceiling (centred) + bedsides + wardrobe driver Inside: 1.5sqmm L + N + E (BR1 ceiling + bedsides + wardrobe driver)
F6	25mm	Riser -> BR1 bedside + study sockets at 600mm FFL Inside: 2.5sqmm L + N + E (bedside L + R + study)
F7	25mm	Riser -> BR1 AC socket at 1850mm FFL Inside: 4sqmm L + N + E (20A AC socket)
F8	25mm	Riser -> T1 geyser + lights bundle Inside: 2.5sqmm L + N + E (T1 geyser 20A + lights)
F9	25mm	Riser -> FF Living ceiling + front balcony + corridor lights Inside: 1.5sqmm L + N + E (FF Living lights + balcony + corridor)
F10	25mm	Riser -> FF Living central wall (router/AP mount) at 2400mm FFL Inside: 2x Cat6 UTP (router + spare) + draw wire Wall plate Cat6 keystone x2 + 5A power socket 300mm beside (on D9 lighting circuit). Equidistant between BR1 and BR2.
F11	25mm	Riser -> BR2 study wall, 700mm FFL Inside: 1x Cat6 UTP + draw wire
F12	25mm	Riser -> BR1 study wall, 700mm FFL Inside: 1x Cat6 UTP + draw wire
F13	16mm (provision)	FF AP -> FF balcony soffit (future outdoor mesh AP) Inside: PULL STRING ONLY - outdoor Cat6 future (balcony AP) Pull draw wire only. Cap at IP67 keystone box at balcony soffit corner.
F14	25mm (provision)	Riser -> front balcony NW corner soffit/parapet (CAM-3) Inside: PULL STRING + 1x outdoor Cat6 - IP67 back-box for ColorVu 4MP Elevated overview of compound, gate, driveway. 5500mm ground level.
F15	25mm (provision)	Riser -> continue UP to terrace level (CAM-5) Inside: PULL STRING + 1x outdoor UV-resistant Cat6 - IP67 back-box at terrace parapet Use UV-rated conduit for last 500mm above roof. Mount on parapet beside terrace door.



Riser bundles (5 power + 3 data + water-auto = 9 conduits up the staircase W wall)

TERRACE (CAM-5 + Sintex JB)

Rule: max 3 circuit wires per 25mm conduit. Use SEPARATE conduits per bundle.

- B1: D5+D6 (BR2 lights/sockets)**
Wire: 2x 1.5/2.5sqmm
- B2: D1+D2 (BR1 lights/sockets)**
Wire: 2x 1.5/2.5sqmm
- B3: D3+D4+D7+D8 (T1+T2 geyser/lights)**
Wire: 2.5sqmm + 1.5sqmm
- B4: D9 (FF corridor + balconies)**
Wire: 1.5sqmm
- B5: E4+E5 (BR1+BR2 AC)**
Wire: 4sqmm
- LV1: R-FF-1 (FF AP, 2x Cat6 + spare)**
Wire: Cat6 UTP
- LV2: R-FF-2 (BR2 study)**
Wire: Cat6 UTP
- LV3: R-FF-3 (BR1 study)**
Wire: Cat6 UTP
- CAM-3 -> FF balcony NW**
Wire: Cat6 outdoor
- CAM-5 -> terrace parapet**
Wire: Cat6 outdoor UV
- C-Sintex-1/2 (water tank + float)**
Wire: Cat6 + 2-core

FIRST FLOOR (FF Living wall + bedrooms)

GROUND FLOOR slab (top)

GROUND FLOOR (DB + niche level)